1. Boiling

Continuous boiling of water for a long time above a certain temperature kills the bacterias. It is the most effective method of disinfection. It is highly impracticable to boil water on large scale for public water supplies. However, during water borne epidemics, it is advisable to boil water before consumption.

2. Excess Lime Treatment

Addition of excess lime to water removes salts and also kills the bacterias. Use of excess lime increases the pH value of water. When the pH of water rises to about 9.5 or so, 99.93 to 100% bacterias are removed even from highly polluted waters.

After disinfection, by adopting some suitable method the excess lime is removed from water before its public supply. Further this method cannot protect water from recontamination.

Activity 3

Collect the data regarding disinfection of water in your locality.

4.6 Water Softening

Water softening is the process of reduction or removal of hardness from water.

4.6.1 Purpose of water softening

1. To reduce soap consumption
2. To reduce corrosion and incrustation of pipes and fittings
3. To improve the taste of food preparations
4. To reduce scaling in boilers
5. To minimise its interference in dyeing systems

4.6.2 Hardness of Water

It is the characteristic which prevents the lathering of soap. It is caused by the presence of certain salts of calcium and magnesium dissolved in water.

Hardness of potable water is 5 to 8 degrees. Hardness less than 5 degrees is tasteless and above 8 degrees produces undesirable effects. (One degree of harness = 14.25 ppm).

Types of Hardness

1. Temporary hardness or Carbonate hardness
2. Permanent hardness or Non-carbonate hardness

Temporary hardness is due to the presence of bicarbonates of calcium and magnesium. Permanent hardness is caused by the presence of sulphates and chlorides of calcium and magnesium.

1. Removal of temporary hardness

The temporary hardness can be removed by boiling or by adding lime which is called as lime process. Lime process is otherwise known as clark process.

The principle involved in this process is the neutralization of carbon-di-oxide with milk of lime.

2. Removal of permanent hardness

The permanent hardness can be removed by special methods of water softening. Any of the following methods can be adopted.

i. Lime-soda process
ii. Zeolite process or Base exchange process
iii. Demineralisation

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